

Vision Transformers Outperform CNNs for Brain Tumour Segmentation in Multi-Site MRI Studies

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Journal: Radiology: Artificial Intelligence, 2025

DOI: 10.1148/ryai.2025230089

ABSTRACT

Accurate segmentation of brain tumours from MRI is critical for treatment planning and outcome prediction. CNN-based architectures have dominated this task, but their limited global context modelling may restrict performance on heterogeneous tumour morphologies. We systematically compare Vision Transformer (ViT) architectures against leading CNN baselines across the BraTS 2024 multi-site benchmark. ViT-based models achieve a mean Dice score of 0.887 for whole tumour segmentation versus 0.862 for the best CNN baseline, a statistically significant improvement ($p < 0.001$). Analysis of failure cases reveals CNNs fail disproportionately on infiltrative tumour boundaries — a clinically consequential region — where ViTs show a 31% reduction in false positive volume.

METHODS

- Dataset: BraTS 2024 — 2,200 multi-parametric MRI scans across 22 sites
- Modalities: T1, T1ce, T2, FLAIR
- Models compared: ViT-B, ViT-L, Swin-UNETR vs nnU-Net, ResU-Net, TransUNet
- Metrics: Dice score, Hausdorff distance (95th percentile), false positive volume
- Statistical testing: Wilcoxon signed-rank test with Bonferroni correction

KEY FINDINGS

1. Swin-UNETR achieves Dice 0.887 vs nnU-Net 0.862 for whole tumour
2. ViT advantage concentrated in infiltrative tumour boundary regions
3. ViTs require 2.3x more GPU memory at inference — a practical constraint
4. Pre-training on large non-medical image dataset (ImageNet-22k) critical for ViT performance
5. No significant difference on small tumours ($<5\text{cm}^3$) — CNNs competitive here

LIMITATIONS

- Computationally expensive — 2.3x memory vs CNNs; limits deployment on standard clinical workstations
- Requires large pre-training dataset to outperform CNNs
- Not validated on paediatric brain tumours
- Study design doesn't address longitudinal tracking (tumour progression)

RELEVANCE TO ML IN CLINICAL DIAGNOSTICS: HIGH

Important architecture comparison for anyone deploying segmentation models clinically. The compute cost finding is a practical constraint that affects deployment decisions.